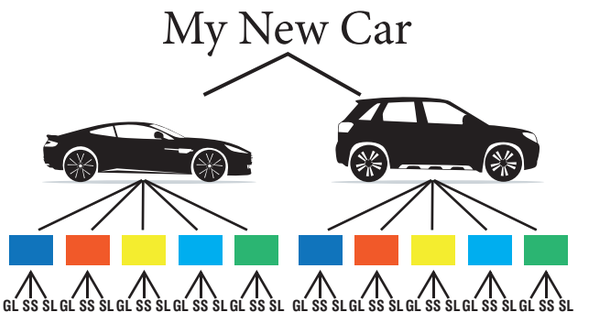


Figure 4.2

We shall now illustrate the different rules described above through examples

Example 4.3 A School library has 75 books on Mathematics, 35 books on Physics. A student can choose only one book. In how many ways a student can choose a book on Mathematics or Physics?

Solution:

(i) A student can choose a Mathematics book in "75" different ways.
(ii) A student can choose a Physics book in "35" different ways.
Hence applying the Rule of Sum, the number of ways a student can choose a book is $75 + 35 = 110$.

Now we shall discuss the problem stated in our introduction.

Example 4.4 If an electricity consumer has the consumer number say 238:110: 29, then describe the linking and count the number of house connections upto the 29th consumer connection linked to the larger capacity transformer number 238 subject to the condition that each smaller capacity transformer can have a maximal consumer link of say 100.

Solution:
The following figure illustrates the electricity distribution network.

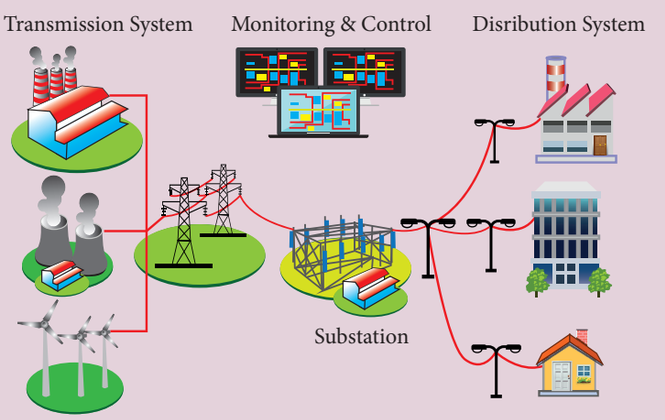


Figure 4.3

There are 110 smaller capacity transformers attached to a larger capacity transformer. As each smaller capacity transformer can be linked with only 100 consumers, we have for the 109 transformers, there will be $109 \times 100 = 10900$ links. For the 110th transformer there are only 29 consumers linked. Hence, the total number of consumers linked to the 238th larger capacity transformer is $10900 + 29 = 10929$.